CS 301 · MEETING 13 · A DEEPER LOOK, 2

Solving by Iterating

A deeper look 2: Kaczmarz, one equation at a time, and when a guess improved step by step beats elimination.

Two ways to solve a system

	direct: elimination	**iterative**
what it does	a fixed recipe, row operations until triangular	a guess, improved one step at a time
what it costs	$O(n^3)$ dense; with fill-in, sparse	a sweep costs the nonzeros; the number of sweeps varies
when it stops	when the recipe ends	when the residual is small enough
what it keeps	the factors L and U, fill included	the matrix and a few vectors
what it gives	the solution to round-off	a solution to the residual you asked for

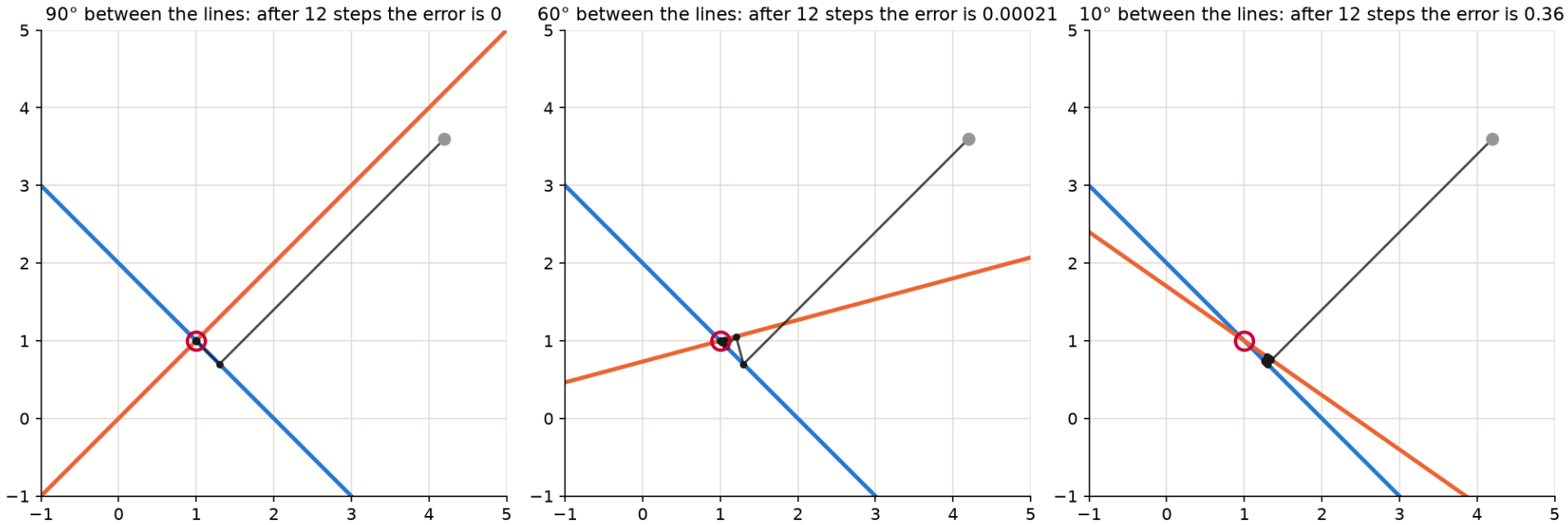
- Both are elimination's rivals in the same game: count, and conditioning

One equation at a time

$$x \leftarrow x - \frac{a_i \cdot x - b_i}{a_i \cdot a_i} a_i$$

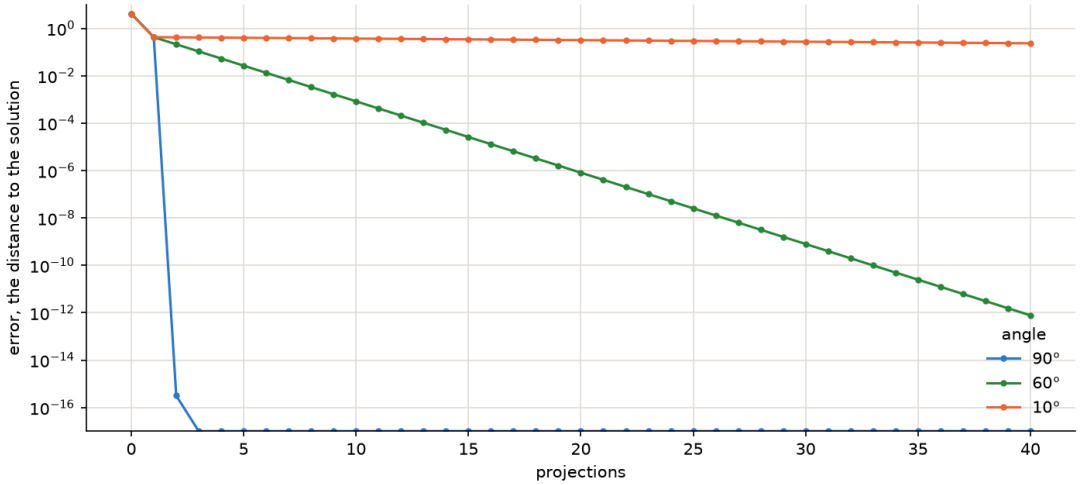
- Row i is a line (a plane): move the guess to its nearest point, down the perpendicular
- After the move, equation i holds exactly; the others, not yet
- Then equation i + 1, and around again: Kaczmarz, 1937

The zigzag, at three angles



- A right angle: two steps and done
- A small angle: the zigzag creeps along the lines

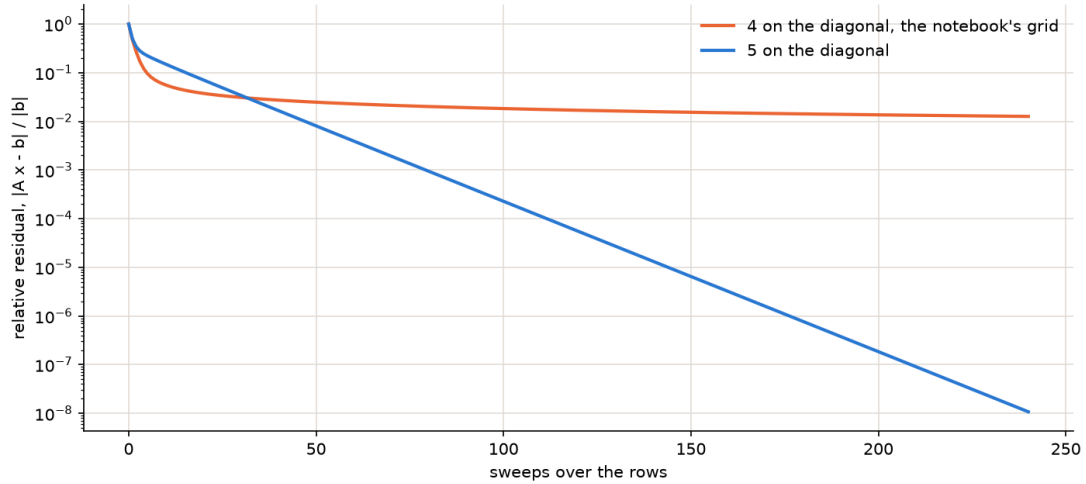
How fast: the angle decides



angle	cos^2 of the angle	error after 2 steps	after 20	after 40
90°	0.000	3.1e-16	0	0
60°	0.250	0.21	8.1e-07	7.7e-13
10°	0.970	0.42	0.32	0.23

- Two lines: each cycle of two projections multiplies the error by $\cos^2\theta$
- Straight lines on a log axis: a fixed factor each cycle
- Nearly parallel, large condition number: a factor near 1, thousands of steps

On a sparse system: a sweep costs the nonzeros




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the 40 x 40 grid: N = 1,600 unknowns, 7,840 nonzeros; one sweep is 7,840 multiply-adds, elimination
in the natural order about 2,560,000
4 on the diagonal, the notebook's grid   residual after 240 sweeps 1.3e-02; not yet below 1e-6
5 on the diagonal                        residual after 240 sweeps 1.1e-08; below 1e-6 after 177
sweeps
```

- One sweep over the rows: one multiply-add for each nonzero
- The number of sweeps is the conditioning again

When to stop: the residual

$$\text{stop when } \frac{\|Ax - b\|}{\|b\|} \leq \text{tolerance}$$

- The residual is what we can compute; it is the stopping rule
- The error is bounded by about the condition number times the residual
- Ask for a smaller residual: more sweeps, the same cost each

Picking the rows at random

- In turn, the rows may be visited in a bad order for a long time
- At random, in proportion to the row's size: expected error factor $1 - 1/\kappa_F^2$ for each projection
- Strohmer and Vershynin, 2009: the randomized Kaczmarz method
- The same idea, rows at random, is how the large models of today are trained

A family of iterations

method	one step	where it is used
Kaczmarz	project onto one row	tomography, signal recovery, huge overdetermined systems
Jacobi, Gauss–Seidel	solve one equation for its own unknown	grids, the walk's system, as a smoother
conjugate gradients	a clever step along the residual	symmetric systems: `scipy.sparse.linalg.cg`

- All of them: a sweep costs O(nnz); the sweeps count follows the conditioning
- None of them creates fill: the matrix is only read

Direct or iterative?

choose	when
elimination, `solve`	the system is small or dense, or many right-hand sides share one matrix
sparse elimination, `spsolve`	the fill stays small, as on a grid with a good order
an iteration	the matrix is huge and sparse, a modest residual is enough, or the fill would not fit

- Both answer the same question of this meeting: how many operations, and how well conditioned